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(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Complete Placement Roadmap

A structured preparation guide covering Aptitude, DSA, Core CS, and AIML

Placement & Interview Preparation Program
Compiled Reference Document



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1. Phase 1 — Aptitude Foundation

A. Quantitative Aptitude

Number systems	Divisibility	HCF & LCM
Simplification	Fractions and decimals	Percentages
Ratio and proportion	Averages	Profit, loss and discount
Simple interest	Compound interest	Partnership
Mixtures and allegations	Time and work	Pipes and cisterns
Time, speed and distance	Problems on trains	Boats and streams
Permutations	Combinations	Probability
Ages	Clocks	Calendars

Data Interpretation

Tables	Bar graphs	Line graphs
Pie charts	Caselets	

B. Logical Reasoning

Number series	Alphabet series	Alphanumeric series
Coding-decoding	Blood relations	Direction sense
Seating arrangement	Linear arrangement	Circular arrangement
Puzzles	Syllogisms	Statements and conclusions
Statements and assumptions	Data sufficiency	Venn diagrams
Analogy	Classification	Odd one out
Missing numbers	Logical deduction	

C. Verbal Ability

Grammar	Parts of speech	Tenses
Subject-verb agreement	Articles	Prepositions
Pronouns	Conjunctions	Sentence correction
Sentence completion	Vocabulary	Synonyms

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Antonyms	One-word substitutions	Idioms and phrases
Para jumbles	Reading comprehension	Error spotting
Fill in the blanks		

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2. Programming Fundamentals

Strongly recommend C/C++ or Java. Students should understand programming concepts rather than only memorising syntax.

Basics

Variables	Data types	Input/output
Operators	Conditional statements	Loops
Functions	Recursion	Scope
Exception handling		

Data Structures in Programming

Arrays	Strings	Lists
Tuples	Sets	Dictionaries / Hash Maps
Stacks	Queues	

Important Programming Problems

Reverse a number	Palindrome	Armstrong number
Prime numbers	Fibonacci	Factorial
GCD/LCM	Counting digits	Sum of digits
Reverse string	Character frequency	Anagram
Duplicate elements	Second largest element	Missing number
Remove duplicates	Matrix operations	

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3. DSA — Very Important

This should probably be the main technical component for 3rd years.

Arrays

Traversal	Searching	Sorting
Two pointer	Sliding window	Prefix sum
Kadane's algorithm		

Searching

Linear search	Binary search	Binary search variations
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Sorting

Bubble sort	Selection sort	Insertion sort
Merge sort	Quick sort	Counting sort

Strings

Palindrome	Anagram	Frequency counting
Substrings	Longest substring	String manipulation

Linked List

Singly linked list	Doubly linked list	Circular linked list
Insert	Delete	Reverse
Find middle	Detect cycle	Merge two linked lists

Stack

Implementation	Parentheses matching	Infix/postfix
Next greater element	Min stack	

Queue

Normal queue	Circular queue	Deque
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Priority queue		
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Trees

Binary tree	BST	Height
Diameter	Lowest common ancestor	

Tree Traversal

Inorder	Preorder	Postorder	Level order
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Heap

Min heap	Max heap	Priority queue
Heap sort		

Graphs

Graph representation	BFS	DFS
Connected components	Cycle detection	Shortest path
Dijkstra	Topological sorting	

Dynamic Programming

Fibonacci DP	0/1 Knapsack	Coin change
Longest common subsequence	Longest increasing subsequence	Grid problems

Backtracking

N-Queens	Subsets	Permutations
Combination problems		

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4. Algorithmic Thinking

Students should learn how to approach a problem. Teach the following sequence:

Step	Action
Step 1	Understand the problem
Step 2	Identify input/output
Step 3	Find brute-force solution
Step 4	Analyse complexity
Step 5	Optimise
Step 6	Code
Step 7	Test edge cases

Complexity

Big O	Time complexity	Space complexity
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$O(1) \rightarrow O(\log n) \rightarrow O(n) \rightarrow O(n \log n) \rightarrow O(n^2) \rightarrow O(2^n)$

Note: This is extremely important for interviews.

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5. Core Computer Science

For 3rd years especially, cover the major CS subjects.

Operating Systems

Process	Thread	Process vs thread
CPU scheduling	FCFS	SJF
Round Robin	Priority scheduling	Context switching
Synchronisation	Deadlock	Memory management
Paging	Segmentation	Virtual memory
Page replacement	File systems	

Important Interview Questions

Process vs thread?
What is deadlock?
Four conditions for deadlock?
What is virtual memory?
What is context switching?

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6. DBMS

Basics

Database	DBMS	RDBMS
Tables	Keys	

SQL

SELECT	WHERE	GROUP BY
HAVING	ORDER BY	DISTINCT
LIMIT	INSERT	UPDATE
DELETE		

Important SQL Concepts

Primary key	Foreign key	Candidate key
Composite key	Subqueries	Aggregate functions
Window functions		

Joins

INNER	LEFT	RIGHT	FULL
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DBMS Theory

Normalisation	1NF	2NF
3NF	BCNF	Transactions
ACID	Indexing	Views
Stored procedures	Triggers	

Note: SQL should have a separate practice session every week.

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7. Computer Networks

Network basics	LAN/WAN/MAN	OSI model
TCP/IP model	IP addressing	IPv4
IPv6	MAC address	DNS
DHCP	HTTP	HTTPS
TCP	UDP	TCP vs UDP
HTTP vs HTTPS	Router	Switch
Hub	Firewall	Ports
Client-server architecture	Socket basics	

Note: Interview questions should be heavily practised here.

8. OOP

Very important because many placement companies ask OOP.

Concepts

Class	Object	Constructor
Encapsulation	Abstraction	Inheritance
Polymorphism		

Advanced

Method overloading	Method overriding	Interface
Abstract class	Access modifiers	Static
Final	Composition	Association

Must-Practise Questions

Overloading vs overriding
Abstraction vs encapsulation
Interface vs abstract class

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Inheritance types

Why OOP?

9. Software Engineering

SDLC	Waterfall	Agile
Scrum	Requirements	Design
Testing	Deployment	Maintenance
Version control	Git	GitHub

Git

git init	git clone	git add
git commit	git push	git pull
branches	merge	pull request

10. AIML Core Topics

Because this is AIML, don't make placement preparation only generic CS.

Mathematics for ML

Linear algebra	Vectors	Matrices
Probability	Statistics	Mean
Median	Variance	Standard deviation
Correlation	Covariance	Bayes theorem

Machine Learning — Supervised Learning

Linear regression	Multiple regression	Logistic regression
KNN	Naive Bayes	Decision tree
Random forest	SVM	Gradient boosting

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Machine Learning — Unsupervised Learning

K-means	Hierarchical clustering	PCA
DBSCAN		

ML Concepts

Training	Validation	Testing
Overfitting	Underfitting	Bias
Variance	Regularisation	Cross-validation
Feature engineering	Feature selection	

Evaluation — Classification

Accuracy	Precision	Recall
F1-score	Confusion matrix	ROC-AUC

Evaluation — Regression

MAE	MSE	RMSE	R^2
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11. Deep Learning

Neural networks	Perceptron	Activation functions
Forward propagation	Backpropagation	Gradient descent
Loss functions	Optimisers	

CNN

Convolution	Filters	Pooling
Feature maps	CNN architecture	

RNN

Sequence data	RNN	LSTM	GRU
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Modern AI

Transformers	Attention	Embeddings
LLM basics	Generative AI	Prompt engineering
RAG basics		

Note: Don't go extremely deep for placement training. Focus on interview-level understanding + implementation.

12. Data Science

NumPy	Pandas	Matplotlib
Data cleaning	Missing values	Outlier detection
Exploratory Data Analysis	Feature engineering	Data visualisation
Basic statistics		

Students should be able to take a CSV and perform:

Load → Clean → Analyse → Visualise → Train → Evaluate

13. Computer Vision

For AIML students:

Image representation	Pixels	RGB/HSV
Image preprocessing	Image filtering	Edge detection
Thresholding	Morphological operations	OpenCV
Object detection	Image classification	CNN
YOLO basics		

14. Generative AI

This is becoming increasingly useful in placements. Cover:

Generative AI basics	LLM	Transformer
Prompt engineering	Zero-shot prompting	Few-shot prompting

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Embeddings	Vector databases	RAG
AI agents	Fine-tuning basics	Hallucination
AI evaluation	Responsible AI	

15. Cloud & DevOps Basics

Placement companies increasingly expect basic awareness.

Cloud

What is cloud computing?	IaaS	PaaS
SaaS	Virtual machines	Containers
Storage	Compute	AWS/Azure/GCP basics

Docker

Container	Image	Dockerfile
Docker Compose	Container vs VM	

CI/CD

GitHub Actions basics	Build	Test	Deploy
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16. Coding Platforms

Train students on:

LeetCode	HackerRank	CodeChef
GeeksforGeeks	Codeforces	

Note: Don't make them solve random problems. Use a structured progression:

Easy → Easy/Medium → Medium → Company-specific

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17. Soft Skills & Placement Readiness

Communication Skills	English & Vocabulary	Self-Introduction
Presentation Skills	Group Discussion (GD)	Leadership & Teamwork
Problem Solving & Critical Thinking	Time Management	Resume Building & ATS
LinkedIn & GitHub Profile	Project Explanation	HR Interview Preparation
Technical Interview Preparation	Mock Interviews & Mock Tests	Professional Etiquette

Cybersecurity Basics

Authentication & Authorisation	Encryption & Hashing	HTTPS
SQL Injection & XSS	Password & Token Security	
